

# Reconstructing frustrated signs from amplitudes with fixed-node and Krylov updates

A. Otaifi<sup>1</sup>

<sup>1</sup>*Arnold Sommerfeld Center for Theoretical Physics,  
Ludwig-Maximilians-Universität München, 80333 München, Germany*

(Dated: October 4, 2026)

We split a real wave function into signs and positive amplitudes,  $\psi = sa$ , and update them alternately in the frustrated square-lattice  $J_1$ - $J_2$  model at  $J_2/J_1 = 0.5$ . The signs follow from a label-free Krylov step,  $s' = s \operatorname{sgn}[T - (H\psi)/\psi]$ ; the amplitude from lattice fixed node, which cannot raise the energy. Given the exact amplitude, three to four Krylov steps recover the exact ground-state signs on a  $4 \times 4$  cluster for  $0.4 \leq J_2/J_1 \leq 1$ . Alternating both updates with exact fixed node converges to the exact ground state on  $4 \times 4$  and 20-site clusters, about three times faster with Anderson acceleration. On  $6 \times 6$  and  $8 \times 8$ , Krylov signs built from a neural-network amplitude reduce the fixed-node energy gap to the network from  $4 \times 10^{-3}$  to  $4\text{--}6 \times 10^{-4}$  per site, without closing it. Iterating further requires learning the fixed-node amplitude: on  $4 \times 4$  a network trained on the frozen fixed-node energy from samples iterates the loop stably, while on  $8 \times 8$  learned updates have not yet lowered the energy.

## I. INTRODUCTION

$$\psi(x) = s(x)a(x), \quad s(x) = \pm 1, \quad a(x) = |\psi(x)| > 0, \quad (1)$$

## II. METHODS

### A. Model and error measures

$$H = J_1 \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + J_2 \sum_{\langle\langle ij \rangle\rangle} \mathbf{S}_i \cdot \mathbf{S}_j, \quad (2)$$

$$w_s = \sum_{x: s(x) \neq s_0(x)} a_0(x)^2, \quad (3)$$

### B. One-step Krylov sign update

$$r_s(x) = \frac{(H\psi)(x)}{\psi(x)}. \quad (4)$$

$$s_K(x) = s(x) \operatorname{sgn}[T - r_s(x)] : \quad (5)$$

### C. Fixed-node amplitude update

### D. Accelerating the amplitude iteration

$$x_{k+1} = g(x_k) - \sum_j \gamma_j \Delta g_j, \quad \gamma = \arg \min_{\gamma} \left\| f_k - \sum_j \gamma_j \Delta f_j \right\|. \quad (6)$$

### E. Learning the amplitude from walkers

$$f(x) \propto a(x)\phi_{\text{FN}}(x). \quad (7)$$

$$a_{\text{new}}^2(x) \propto a(x)\phi_{\text{FN}}(x), \quad (8)$$

$$\delta\theta = (S + \lambda I)^{-1}g, \quad (9)$$

$$L(\theta) = \frac{\langle a_\theta s_k | H_{\text{FN}}[a_k, s_k] | a_\theta s_k \rangle}{\langle a_\theta | a_\theta \rangle} \quad (10)$$

### F. Numerical setup

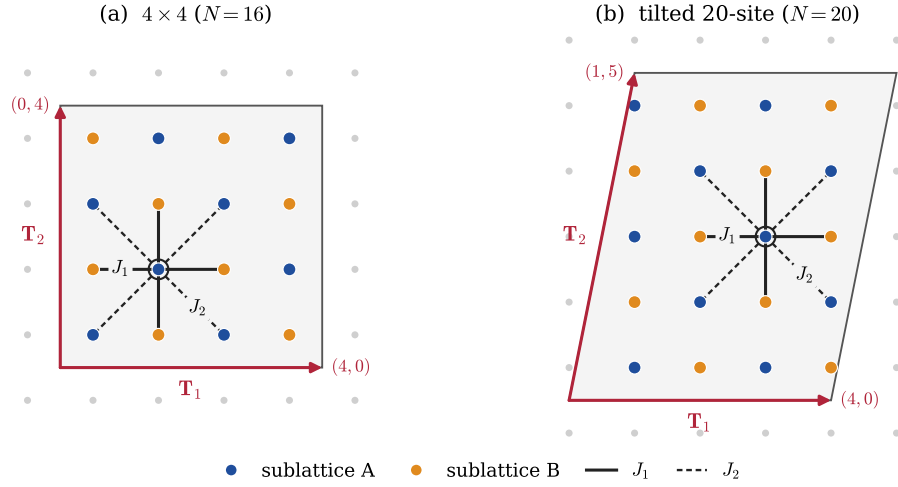


FIG. 1. Exact clusters with periodic boundaries: (a)  $4 \times 4$ , (b) tilted 20-site torus. Colours: Marshall sublattices; arrows: periodic vectors;  $J_1/J_2$  bonds shown for one site.

### III. RESULTS

#### A. Krylov sign update

##### 1. Exact amplitude

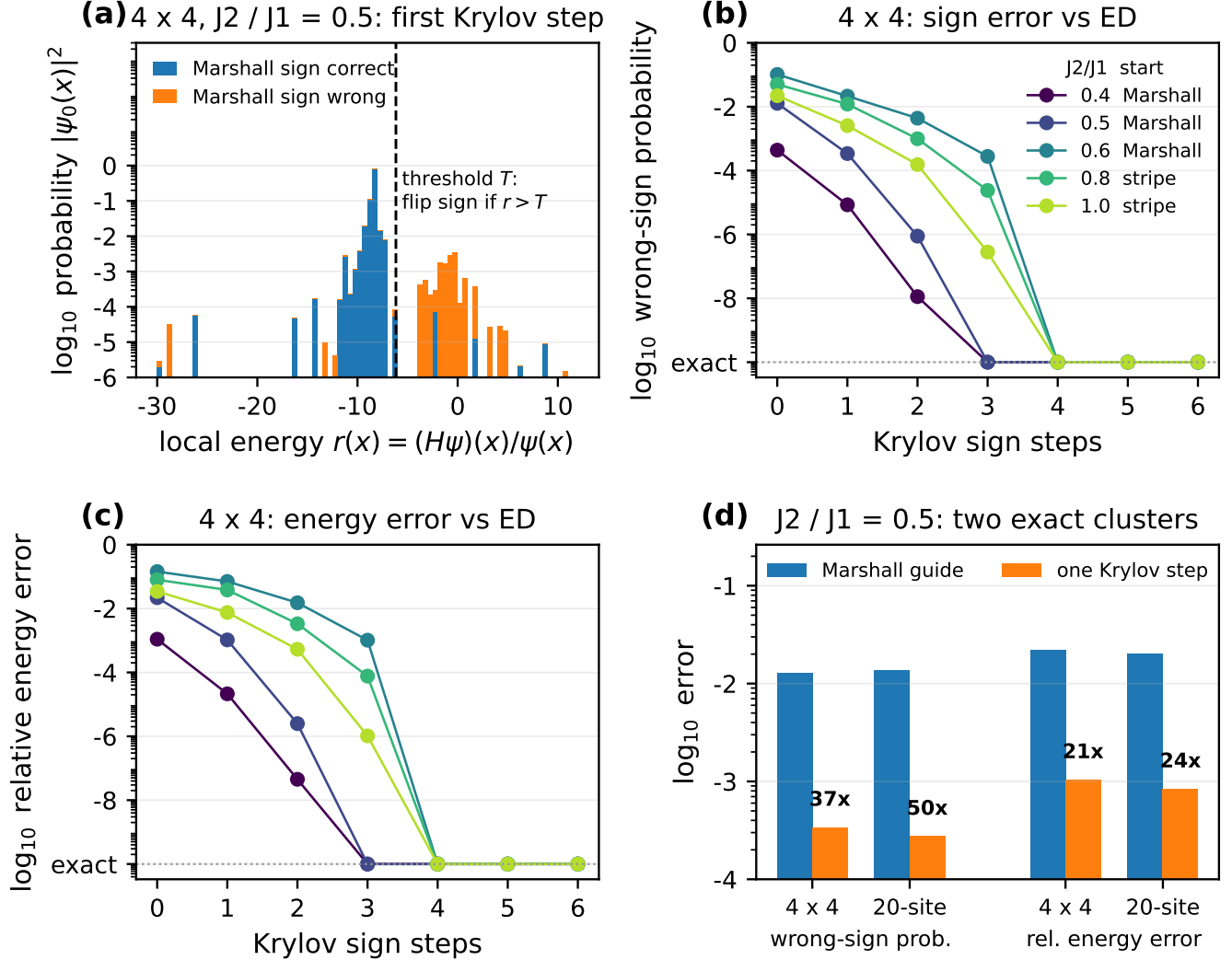


FIG. 2. Sign correction by Krylov steps, starting from Marshall signs. Only the signs change; the amplitude stays the exact ED one. (a)  $r(x)$  at  $J_2/J_1 = 0.5$ ; dashed line: threshold  $T$ . (b) Wrong-sign probability and (c) energy error vs number of Krylov sign steps, compared with ED. (d) First step on the  $4 \times 4$  and 20-site clusters (Fig. 1).

## 2. Approximate amplitude

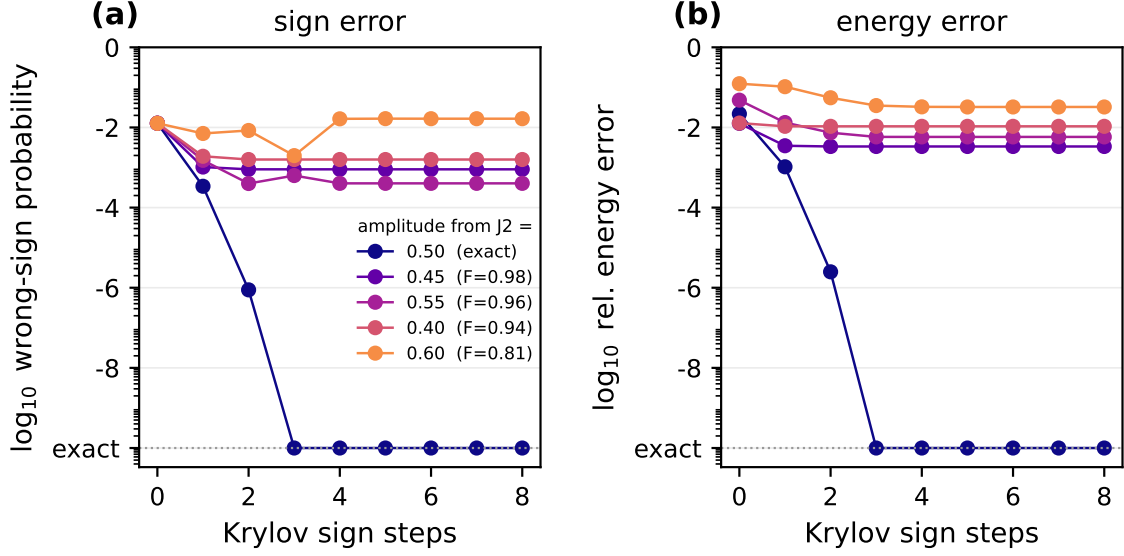


FIG. 3. Krylov sign steps on an approximate amplitude at  $J_2/J_1 = 0.5$ ; the amplitude is the exact one of  $J'_2$  (fidelity  $F$ ). (a) Sign and (b) energy error vs number of Krylov sign steps, compared with ED.

## 3. Neural-network amplitude on $6 \times 6$

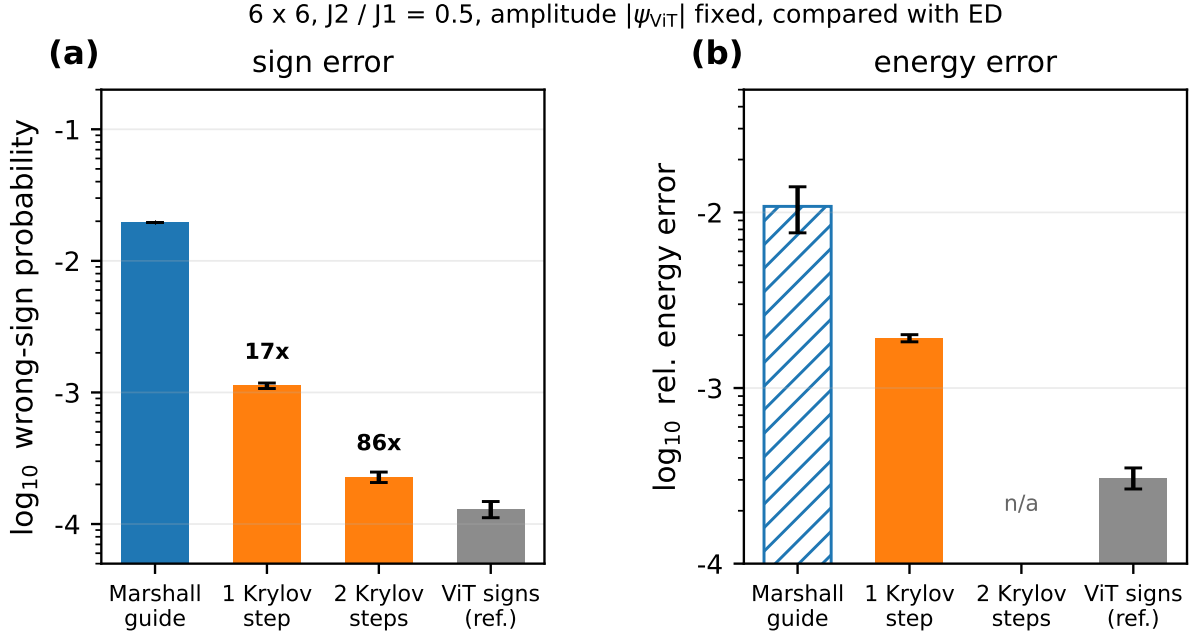


FIG. 4. Krylov sign steps on the ViT amplitude ( $6 \times 6$ ), compared with ED, with the ViT's own signs as reference. (a) Wrong-sign probability ( $4 \times 10^5$  exact samples). (b) Relative energy error; the Marshall value (hatched) is a lower bound from a smaller sample, the two-step energy is not resolved.

### B. Fixed-node-Krylov loop

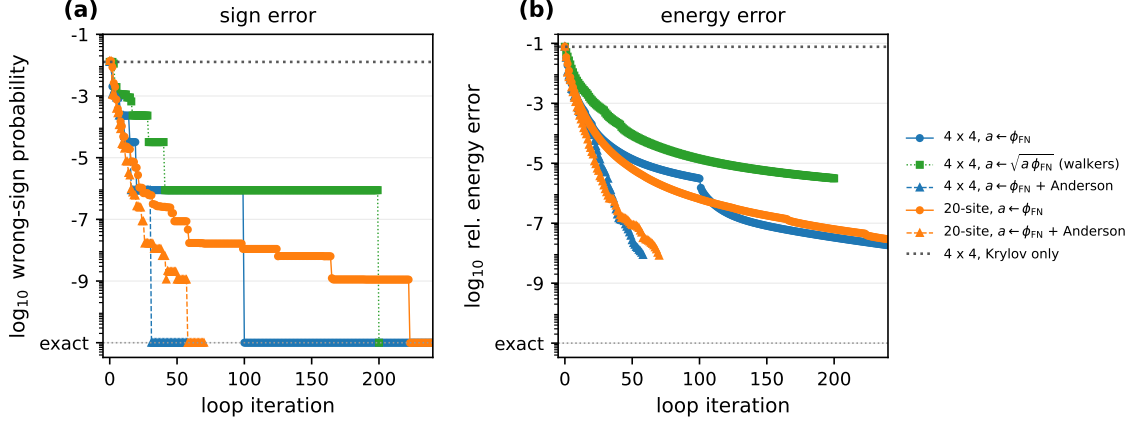


FIG. 5. Fixed-node-Krylov loop with exact diagonalization, started from the  $J_2 = 0$  amplitude with Marshall signs, for three amplitude updates:  $a \leftarrow \phi_{FN}$ ;  $a \leftarrow \sqrt{a} \phi_{FN}$ , as learned from walkers; and  $a \leftarrow \phi_{FN}$  with Anderson extrapolation from the last three iterates. Dotted: Krylov steps alone from the same start, which change no sign. (a) Sign and (b) energy error vs iteration, compared with ED. All values are exact (no statistical error).

### C. Sampled fixed node on $4 \times 4$

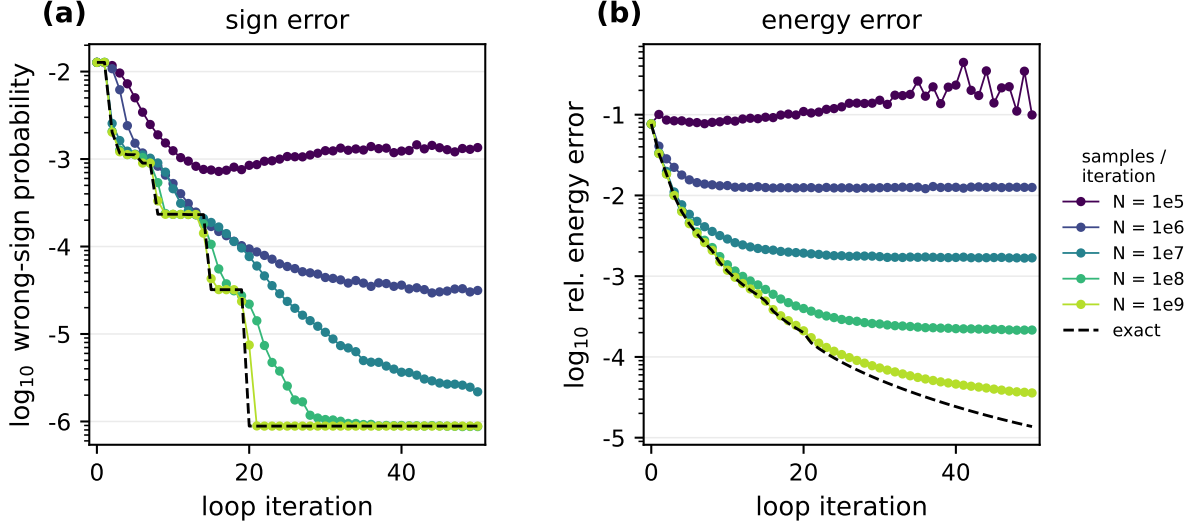


FIG. 6. Fixed-node-Krylov loop on  $4 \times 4$  with  $\phi_{FN}$  estimated from  $N$  independent samples per iteration (mean over four seeds; two for  $N = 10^9$ ), and the exact loop. (a) Sign and (b) energy error vs iteration, compared with ED.

### D. First iteration at scale: $6 \times 6$ and $8 \times 8$

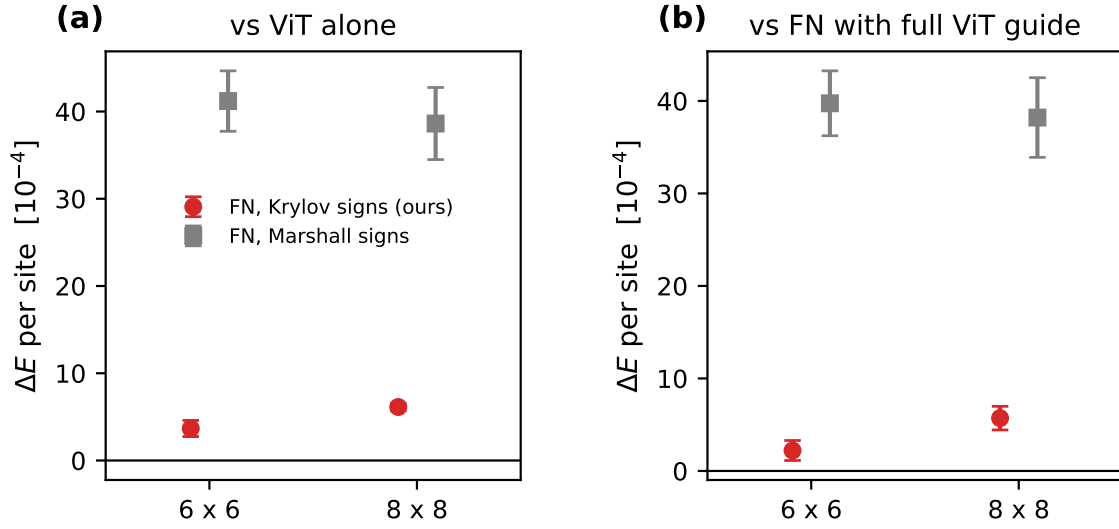


FIG. 7. One untrained iteration on  $6 \times 6$  and  $8 \times 8$ : fixed-node energy with Krylov or Marshall signs on the ViT amplitude, relative to (a) the ViT variational energy and (b) fixed node with the full ViT guide.

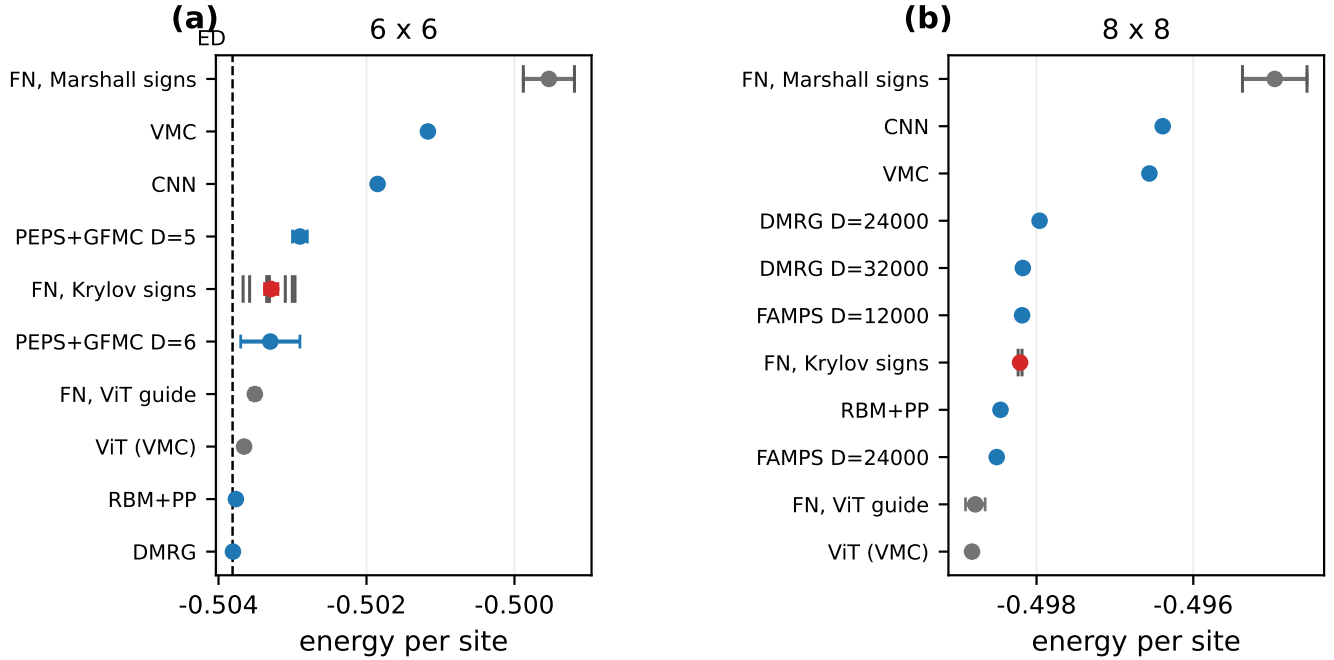


FIG. 8. Energies per site at  $J_2/J_1 = 0.5$ : our fixed-node values (Marshall, Krylov, full ViT guide;  $M = 128$ , not extrapolated), the ViT, and published results: (a)  $6 \times 6$ , (b)  $8 \times 8$ . ED [1], VMC [2], CNN [3], DMRG [4], RBM+PP [5], PEPS+GFMC [6], FAMPS [7].

### E. Closing the loop: learning the amplitude

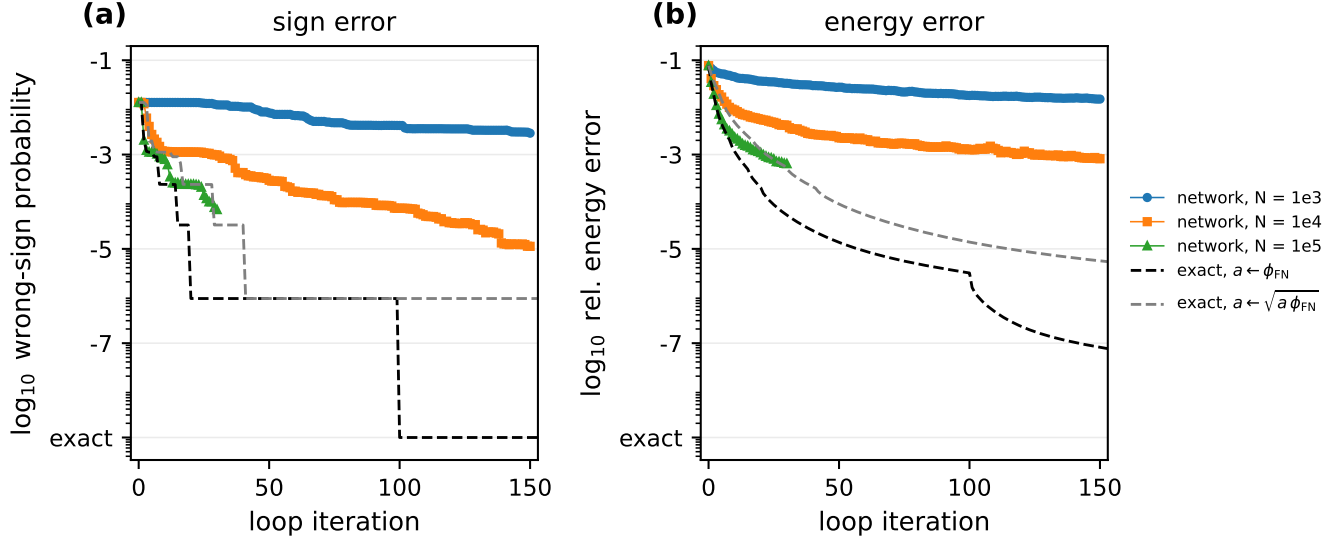


FIG. 9. Fixed-node-Krylov loop on  $4 \times 4$  with a network amplitude trained on the frozen fixed-node energy from  $N$  samples per SR step (mean over three seeds), compared with the exact loops. (a) Sign and (b) energy error vs iteration, compared with ED.

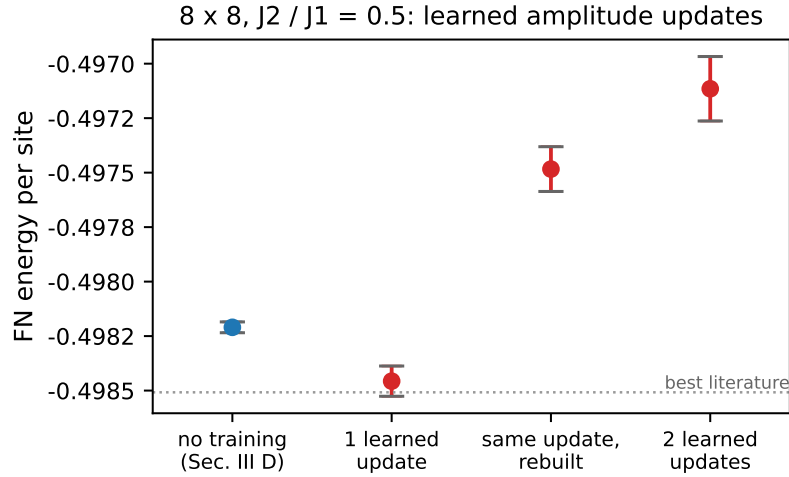


FIG. 10.  $8 \times 8$  fixed-node energy per site along learned amplitude updates (two walker populations each). Dotted: best same-geometry literature value [7].

## IV. DISCUSSION

### Appendix A: Direct second Krylov step

### Appendix B: Statistical conventions

- 
- [1] H. J. Schulz, T. A. L. Ziman, and D. Poilblanc, Magnetic order and disorder in the frustrated quantum Heisenberg antiferromagnet in two dimensions, *J. Phys. I France* **6**, 675 (1996).
  - [2] W.-J. Hu, F. Becca, A. Parola, and S. Sorella, Direct evidence for a gapless  $Z_2$  spin liquid by frustrating Néel antiferromagnetism, *Phys. Rev. B* **88**, 060402(R) (2013).
  - [3] K. Choo, T. Neupert, and G. Carleo, Two-dimensional frustrated  $J_1$ - $J_2$  model studied with neural network quantum states, *Phys. Rev. B* **100**, 125124 (2019).
  - [4] S.-S. Gong, W. Zhu, D. N. Sheng, O. I. Motrunich, and M. P. A. Fisher, Plaquette ordered phase and quantum phase diagram in the spin-1/2  $J_1$ - $J_2$  square Heisenberg model, *Phys. Rev. Lett.* **113**, 027201 (2014).
  - [5] Y. Nomura and M. Imada, Dirac-type nodal spin liquid revealed by refined quantum many-body solver using neural-network wave function, correlation ratio, and level spectroscopy, *Phys. Rev. X* **11**, 031034 (2021).
  - [6] H.-Y. Lin, Y. Guo, R.-Q. He, Z. Y. Xie, and Z.-Y. Lu, Green's function Monte Carlo combined with projected entangled pair state approach to the frustrated  $J_1$ - $J_2$  Heisenberg model, *Phys. Rev. B* **109**, 235133 (2024).
  - [7] X. Qian and M. Qin, Augmenting density matrix renormalization group with disentanglers, *Chin. Phys. Lett.* **40**, 057102 (2023).